Measuring and Modelling the Nuances of Non-Equilibrium Condensation

Minjae Kim

Supervisor: Dr Teng Cao

Department of Mechanical Engineering

Imperial College London

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Abstract

Non equilibrium condensation of gases such as water and CO_2 is critical to understand in a wide range of engineering applications such as compressors to ejectors, due to its contribution to damages and efficiency losses. This review evaluates computational methods for simulating NEC, comparing Eulerian-Eulerian and Eulerian-Lagrangian frameworks, the impact and methods of representing polydispersion, with regards to computational performance and accuracy. Moreover, experimental techniques are challenged, including pressure distribution and optical droplet sizing via Rayleigh scattering in nozzles and blade cascades. Significant data uncertainties from effects such as boundary layer blockage are identified and emerging methods to model non-equilibrium condensation are indicated.

Statement of Project Objectives

- Bring an overview of the modifications made in classical nucleation theory for non-equilibrium condensation and droplet growth models. Focus on the why the modifications are applied.
- Investigate methods for non-equilibrium condensation experiments, particularly steam flows in nozzles and blade cascades.
- Delve into simulation techniques ranging from numerical methods to flow field definitions.
- Suggest guidelines for simulating non-equilibrium condensation, in modelling and experimentation.

Citation Style

APA 7th Edition will be used due to its guarantee on mentioning DOIs in the bibliography unless it is described alternatively, such as from a university website.

The Use of Artificial Intelligence

Large language models have been used to identify grammar mistakes and improve the flow of the literature review. Furthermore, it has been used to assist in understanding concepts, however it has not been used to write the content or to formulate the arguments presented in this review.

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Introduction

From cloud formation to air conditioning, the phenomenon of condensation is ubiquitous. In the context of turbomachinery, condensation was of interest to understand the behaviour of steam power cycles, where the nucleation and growth of water droplets within the machinery triggers water droplet erosion mechanisms and energy losses. It is estimated that turbine related failures contribute 1 billion dollars per annum to losses in profit within the global utility industry (McCloskey, 2009). Interest in this field has accelerated due to the potential application of supercritical CO_2 (sCO_2) Brayton cycles in power generation, where it shows promise to become a next generation energy extraction method, due to its physical compactness and high efficiency (Ahn et al., 2015; Guo et al., 2022). Condensation is a key concern in the compressor, where without the appropriate countermeasures, such as increasing the compressor temperature beyond the room temperature, pseudo-isentropic expansion occurs with the leading edge of the compressor blade. This results in CO_2 droplet formation, contributing to damage and losses (Lettieri et al., 2017). Moreover, condensation affects the operation characteristics of steam ejectors, carbon capture technology and compressed air storage; the need to understand condensation in industry is clear (G. Zhang, Zuo, et al., 2025).

In the aforementioned applications, homogenous Non-Equilibrium Condensation (NEC) occurs, where the working fluid does not condense at once upon saturation. Instead, it continues to supercool until reaching the Wilson line(Bakhtar et al., 2005; Lettieri et al., 2017). This phenomenon relies on homogenous nucleation that occurs without preexisting surfaces, as opposed to heterogenous nucleation, which requires a surface to initiate condensation.

NEC intensity is governed by droplet nucleation and droplet growth, where both are inherently coupled. However, if nucleation happens as within a short period relative to the droplet growth, it could be assumed that separate equations govern each phenomenon. In tandem with commercial Computational Fluid Dynamics (CFD)

software such as Ansys Fluent or OpenFOAM, dual phase flows can be simulated, which can be critiqued with extensive experimental measurements in literature, ranging from Converging-Diverging (ConDi) Nozzles to blade cascades (Edathol et al., 2020; Hu et al., 2024; Starzmann et al., 2018). Data measurements such as pressure distribution and droplet radius have also been measured, using various methods such as Rayleigh Scattering (Wang et al., 1998).

This review examines various nucleation and droplet growth models, focusing on how they address the underlying characteristics of NEC, such as polydispersity and two-phase slippage. It also identifies areas requiring further investigation, notably droplet agglomeration, where the validity of each modification is evaluated against existing literature. Furthermore, the experimental methods used for validation are critically scrutinised, specifically regarding their approaches to pressure measurement, droplet characterisation and environmental geometry.

Modelling

Classical Nucleation Theory

The thermodynamic force that drives condensation is the chemical potential difference of a given state. Given that the system is not near the critical temperature T_c, and the liquid is considered incompressible, it can be expressed as Equation 1, where k_B is Boltzmann's Constant. T is the temperature of the state, $\Delta\mu$ is the chemical potential difference, and S is the supersaturation ratio. S can be approximated as the ratio between the vapour pressure and saturation pressure, shown in Equation 2, if the vapour acts as an ideal gas.

$$\Delta\mu = k_B T \, ln \, S \tag{1}$$

$$S = \frac{p_v}{p_s} \tag{2}$$

S represents the degree of metastability of a given system, where it is at a local minimum of free energy that is stable to small fluctuations.

Becker, Döring, Volmer, Weber and Zeldovich established and utilised the capillarity approximation which states: a cluster made of n molecules is considered a large homogenous sphere, the liquid and vapour of a system has individual bulk properties, the liquid is incompressible and the surface energy of a given cluster is determined with the planar interfacial tension σ_∞ at T, and the surface area of the cluster, $A(n)$. Their work ultimately results in Equation 3, which relates the nucleation barrier, represented by the critical change in Gibb's free energy ΔG^*, and the minimum number of molecules to form a cluster, n_c. Assuming the droplet is spherical and using the Kelvin relation, an alternative expression for Equation 3 can be developed in the form of Equation 4, where ρ_L is the density of the liquid, R is the ideal gas constant and T_V is the temperature of the vapour.

$$\Delta G_* = \frac{1}{3}\sigma_\infty A(n_*) \tag{3}$$

$$\Delta G_* = \frac{16\pi\sigma_\infty^3}{3(\rho_L R T_V \ln S)^2} \tag{4}$$

In CNT, cluster growth is assumed to be dominated by the addition of monomers, which is the gaseous unit molecule of the unary system. Moreover, when a monomer collides with a cluster, it is assumed that the absorption of the monomer is guaranteed and there is independence between successive events that affect the number of a particles in a cluster, thus could be represented as a Markov process (Kalikmanov, 2013a). This results in Equation 5, which outputs J, the steady state nucleation rate per unit mass of vapour. It is good to note that in CNT, steady state nucleation is assumed to be reached rapidly, where J is the universal rate of growth of any cluster size in a system. J represents solely the rate of production of nucleating seeds, and the development of the droplet is governed by a droplet growth equation. The pre-exponential factor shown in Equation 6, is given by ν, the impingement rate per unit area, thus $\nu A(n_*)$ shows the rate of monomer attachment for a critical cluster. Z, the Zeldovich factor, gives the likelihood of a critical cluster to overcome the energy barrier. With vapour density ρ_v, it computes the base steady state nucleation rate, J_0.

$$J = J_0 e^{-\frac{\Delta G_*}{k_B T}} \tag{5}$$

$$J_0 = Z\nu A(n_*)\rho_v \tag{6}$$

Using Becker and Döring's kinetic theory (Becker & Döring, 1935), Bakhtar formulated Equation 7, which is known as the Becker-Döring formulation. q_c is the condensation coefficient, ρ_{vS} is the saturated vapour density, ρ_l is the liquid density, m is the mass of a single vapour molecule and T_v is the temperature of the vapour (Bakhtar et al., 2005).

$$J_{BD} = q_c S \frac{\rho_{vS}}{\rho_L} \sqrt{\frac{2\sigma_\infty}{\pi m^3}} e^{-\frac{4\pi r_*^2 \sigma_\infty}{3k_B T_v}} \tag{7}$$

Kinetic and Thermodynamic Corrections to Classical Nucleation Theory

Various attempts to tackle the limitations of CNT have been made. Courtney (Courtney, 1961) noted that the partial pressure effects have been omitted on the chemical potential that drives nucleation according to CNT. J modified by Courtney's correction, denoted as J_{CNY} is shown to be J_{BD} divided by S.

Kantrowitz corrected steady state nucleation rate (J_{KANT}) challenges the assumption of J_{BD}, where the formation of droplets occurs in an isothermal condition. In reality the formation of droplets involves the release of latent heat, consequently heating up the droplet itself, resulting in the droplet evaporating, lagging the generation of the critical sized cluster (Kantrowitz, 1951).

Wölk and Strey (Wölk et al., 2002) developed an empirical model for J (J_{WS}), based on experimental measurements of J in a nucleation pulse chamber. J_{WS} employs a temperature dependent exponential function that corrects J_{BD}. The use of this correction is justified based on work done by Oxtoby et al (Zeng & Oxtoby, 1991), where curvature effects of small nuclei affecting surface tension (σ) and spinodal effects are omitted. The spinodal represents a point of thermodynamic instability where the free energy barrier of cluster formation vanishes, leading to rapid growths of both phases. This can be achieved by high S or high amounts of supercooling. However, it is inconsistent in CNT, where the nucleation barrier remains finite in all values of S, as shown in Equation 4. Moreover, as the system approaches close to the spinodal, called the pseudo-spinodal, the effects of stochastic thermal fluctuations determine the

triggering of spinodal decomposition (Kalikmanov, 2013b). σ effects lowers J compared to the expectation, while spinodal effects increases it. This is further supported by Density Functional Theory (DFT) simulations. DFT goes beyond the capillarity approximation of CNT, where the droplet is not considered to have bulk properties, but as a medium with varying density (ρ) and the free energy of the droplet is assumed to vary with ρ. The exponential factor attempts to encapsulate these competing effects. Hale (Hale, 2005) created scaled model similar to J_{WS} (J_{HA}), where the exponential fitting function is proved to be applicable in compounds beyond water, such as nonane and toluene.

Surface Tension Corrections

As seen in equation 4, J is sensitive to the change in surface tension due to its presence in the power of the exponential. Therefore, it is paramount to capture the changes in σ_∞, depending on various conditions. The increase in T reduces σ_∞, due to the increase in kinetic energy of the molecules decreasing the cohesive force that determines the surface boundary of the liquid. This relationship can be modelled by interpolating empirical data from sources such as the National Institute of Standards and Technology REFPROP database, in tandem with thermophysical approaches (Jiang et al., 2017; Mulero et al., 2012).

For small droplet sizes, the curvature of the droplet is non negligible, therefore actual σ values deviate from σ_∞. Methods such as Benson-Shuttleworth (BS) model or Tolman's correction exist, which are grounded in analysis. The BS model captures geometric effects of a cluster when it gets into the molecular level. Molecules have less bonds to break, contributing to the decrease in the average surface energy of the cluster (Benson & Shuttleworth, 1951). Tolman's correction realises that in small droplets, the interface that defines the distinction between two phases in CNT no longer can be considered infinitely thin, and an equimolar surface and a surface of tension must be separately established, resulting in lower σ. Using either geometrically, or thermodynamically informed interpretations of σ, in tandem with an empirical trend of

σ with T, a function for σ can be defined using T and r, the droplet radius. Ultimately σ can be directly substituted instead of σ_∞ into models of J to capture this trend.

Zhang et al (G. Zhang et al., 2020) applied this method to model NEC in a steam ejector, which mixes a relatively low-pressure secondary flow with a motive primary flow, in this case steam, to transfer heat and momentum to enable transport of the secondary flow. In the study, various droplet growth models in tandem with modifications of the nucleation models were compared, where J_{BD} changed with σ corrections and the Hill model was ultimately determined to be the most accurate based on droplet size and pressure measurements.

Droplet Growth Models

While J models the nucleation rate of the critical cluster, a Droplet Growth Model (DGM) represents how the critical cluster evolves in size. There are various types of DGMs, based on the different theories on droplet growth, each defined by its own key assumptions.

A DGM can be formulated in various approaches. It can be understood that the increase in r is controlled by two criteria: there must be mass transfer of a monomer that can be fused with a droplet, and the droplet temperature(T_d) needs to be below T_v. Therefore, in order for sustained condensation, the latent heat caused by droplet growth needs to be conducted away from the droplet and mass diffusion from the bulk vapour should supply monomers to the droplet (Dong et al., 2024).

Gyarmathy(Gyarmathy, 1962) considered a spherical droplet surrounded by a continuum, dominated by the heat conduction criteria. The release in latent heat is due to the vapour monomer being absorbed by the droplet can be used to derive Equation 8, where T_s is the vapour saturation temperature, r is the droplet radius, λ_v is the thermal conductivity of the vapour and L_{vl} is the latent heat of vaporisation.

$$\frac{dr}{dt} = \frac{\lambda_v \left(1 - \frac{r_*}{r}\right)(T_s - T_v)}{L_{vl}\rho_l r} f(\mathrm{Kn}) \tag{8}$$

Importantly, $f(Kn)$ is a correction factor to model NEC beyond a continuum. Kn is the Knudsen number, which represents relative magnitude of a particle's mean free path compared to its characteristic length. The mean free path is the average distance the particle travels before its property is changed due to an interaction. In this context, the property that is focused on is before a droplet's radius is altered due to colliding with a gaseous monomer. Thus, a high Kn represents the droplet in a free molecular regime, while a relatively low Kn represents the droplet in a continuum regime. It is good to note that these regimes are arbitrary, however in literature, $Kn < 0.01$ is the continuum regime, $0.01 < Kn < 0.1$ is in the slip flow regime, $0.1 < Kn < 10$ is the transitional regime and $\mathrm{Kn} > 10$ is considered the free molecular regime (Petruccelli et al., 2024). Furthermore, being able to use a continuum based model then correcting it based on Kn gives the convenience to use readily available methods such as CFD to analyse a system is typically simulated in a continuum. Heat transfer is governed by particles act as a medium and as Kn gets larger, no longer can the droplet be considered as being surrounded by a continuous supply of monomers. Several individuals modified Gyarmathy's approach by proposing their own $f(Kn)$, notably the Young Model (YO). Young's DGM is based on Langmuir's three-layer theory, which divides the flow field surrounding into a bulk continuum-controlled field, an inner Knudsen layer governed by kinetics, and the droplet itself, as shown in Figure 1 (Young, 1993).

Kinetic formulations of DGM consider an alternative view on looking at droplet growth, where the interaction between monomers and the droplet surface dominates the phenomena. Classically, the Hertz-Knudsen Model (HK) considers the mass flux balance, on the rate of evaporation of the droplet and the rate of impingement of a monomer using bulk properties (Ytrehus & Østmo, 1996). The Hill Model (HI) additionally considers the reflective molecules from the arriving and leaving vapour (Hill, 1966). Finally, the Puzyrewski–Król Model (PK) considers the negative feedback mechanism of droplet growth, where the latent heat generated by droplet growth retards the droplet growth itself, by increasing the equilibrium vapour pressure on the

droplet surface (*Numerical Analysis of Hertz-Knudsen Model of Condensation upon Small Droplets in Water Vapor: Instytut Maszyn Przepływowych PAN*, n.d.).

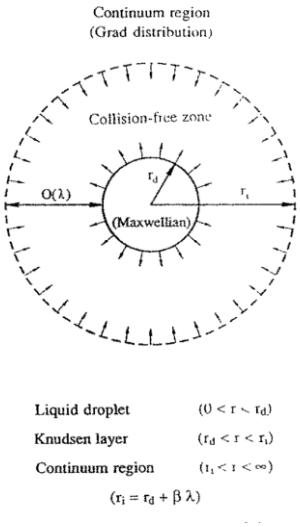


Figure 1: Visual represenation of Langmuir's three-layer theory (Young, 1993).

The inconsistency between the conditions of continuum and the free molecular regime can be attempted to be addressed using hybrid models. Wiśniewski et al considered using a blend function to combine a continuum based DGM and a kinetic based DGM to cover all Kn (Wiśniewski et al., 2022).

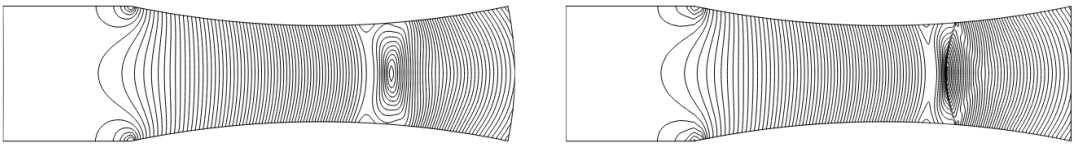
Computational Methods

A binary system of liquid and vapour exists in NEC. Using simulation tools such as Ansys Fluent or OpenFOAM, NEC can be simulated in a 2D or 3D context. Initially NEC in nozzles were analysed in a 1D context along the centreline of the nozzle. However, by accommodating more dimensions, this can take account of unsteady and spatial effects. A Eulerian and Lagrangian frame of reference can be used to simulate the system. Typically, the vapour is modelled in a Eulerian basis. However, droplets are modelled using either reference (Edathol et al., 2020; Fürst et al., 2021a; Hu et al., 2024; Lakzian et al., 2023; Noori Rahim Abadi et al., 2015; Starzmann et al., 2018; G. Zhang, Zhang, et al., 2025; G. Zhang, Zuo, et al., 2025), therefore two approaches are used; a Eulerian-Eulerian (E-E) framework or a Eulerian-Lagrangian (E-L) framework.

Although it is a binary system, the simulation can either use a two phase (2P) or a single-phase approach (1P). An explicit definition for each phases governing heat, mass

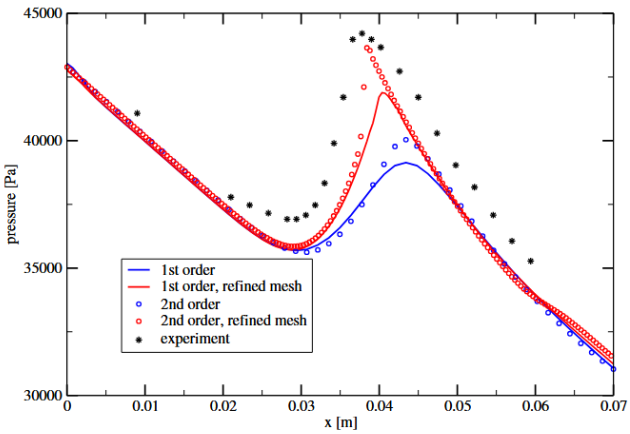
and momentum for a 2P approach gives the potential to simulate more complex phenomena unlike 1P, where properties are averaged using mixture relations originating from wetness distribution (Gerber, 2002). For simplicity, models can implement a no slip condition between the two phases. However, 2P models can introduce a drag force in the governing equations of flow, defined using the Navier-Stokes equation further matching the reality of NEC (Dykas & Wróblewski, 2011; Noori Rahim Abadi et al., 2015).

Changing different elements of a numerical method, such as time integration methods, the order of method and discretisation size impact NEC simulation results. It is shown that mesh refinement in zones of abrupt pressure rises is particularly important to capture NEC effects, as shown in Figure 3 whereby refining the mesh as shown in Figure 2, significant information regarding pressure inside the nozzle due to NEC can be captured(Halama, 2018). This is a particularly an issue for E-E frameworks, in tandem with the fact that the location of the nucleation zone is unknown a priori while in E-L methods, this is not an issue (Fürst et al., 2021b).



(a) 2nd order method (b) 2nd order method, refined grid

Figure 2: Mesh refinement shown in b to improve capturing NEC phenomena compared to a (Halama, 2018).



(a) pressure along the nozzle axis (zoom), experiment [9]

Figure 3: The effect of order of numerical method and mesh refinement in pressure modelling in nozzle flows (Halama, 2018).

Polydisperse samples represent a sample with variation of r unlike monodisperse samples. Secondary nucleation may occur due to expanding already wet steam and unsteady flow regimes. Information regarding smaller r droplets generated by secondary nucleation compared to mature droplets are hindered due to averaging practices of liquid mass fraction and r, commonly deployed with E-E systems could prevent them affecting simulation results (White & Hounslow, 2000). Moreover, knowing droplet distribution is of interest due to potential to be used to represent machinery damage more accurately. However, tracking each particle comes with a higher computational cost. Young (Young, 1992) used bins to reduce computation cost, verifying with nozzle data and turbine cascades.

Complexities arise with interpolating between different grids to extract fluid properties and with streamline tracking and with unsteady 3D flows. Therefore, a solely Eulerian framework could be easier to simulate these scenarios. The Method of Moments (MoM) in tandem with an E-E approach can capture polydispersity. The ith moment of droplet size distribution, M_i can be defined as Equation 9, where f is the number density function.

$$\mathrm{M}_i = \int_0^\infty r^i f \, dr \tag{9}$$

The first 4 moments i.e M_0 to M_3, needs to be calculated per grid cell to be used for fluid dynamics simulations, each representing: mass-based number density, sum of all droplets radii per unit mass, correlation between total droplet surface area and correlation between volume respectively. This contrasts with requiring an arbitrary number of bins in E-L methods depending on the variation of droplet size, giving a computational advantage. Higher order moments can also be used to bring statistical significance such as kurtosis (White & Hounslow, 2000). A notable modification of MoM is Quadrature MoM (QMoM). Using MoM, f needs to be approximated by asserting assumptions such as a linear droplet growth rate. Using Gaussian Quadrature (GQ), M_i can be approximated as Equation 10, where n is the GQ degree and ω_k is the corresponding GQ constants. Afzlifar et al (Afzalifar et al., 2017) claims that QMoM

is more accurate compared to MoM, justifying that statement by showing QMoM matching E-L simulation data from the Moore A nozzle serving as a benchmark.

$$M_i = \sum_{k=1}^{n} r_k^i \omega_k \tag{10}$$

Zhang et al (G. Zhang, Zhang, et al., 2025) presented physical explanation of why polydisperse models match reality better than monodisperse models such as flow deceleration patterns and observed condensation onset by analysing enthalpy-entropy diagrams and concluding that polydisperse droplet populations have a thermal and inertial relaxation originating from gradual buildup of the droplet size distribution compared to monodisperse populations that react synchronously.

Experimental Methods and its Effect on Result Analysis

As shown in Figure 4, pressure measurements and radius measurements are important in validating model accuracy. A key indicator of NEC is the pressure increase as shown in the highlighted area. Typically, nucleation and droplet growth due to NEC happens in a short amount of time. The latent heat produced by the droplet growth abruptly increases the surrounding gases pressure, temperature which declines eventually. This manifests in the phenomenon of 'condensation shocks'. The jump in pressure is a key indicator of condensation shock, as shown in Figure 4. It is observed in turbine cascades and its importance to characterise comes from the conditions deviating from the turbines operating point leading to irreversible losses and influences the fishtail shockwave pattern observed, as shown in Figure 5 (Nagib Elmekawy & Ali, 2020; Yao et al., 2025).

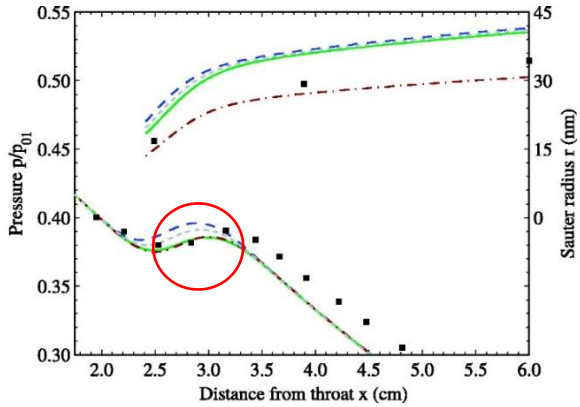


Figure 4: Generic graph of comparing models with experimental nozzle data shown as black dots. Pressure increase is highlighted as a red circle (Starzmann et al., 2018).

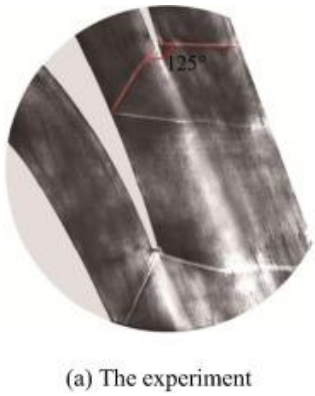


Figure 5: Fishtail shockwaves visualised in blade cascades (Nagib Elmekawy & Ali, 2020).

Droplet Size Measurements

The Sauter mean droplet radius ($\overline{r}_{saut}$) is used for measurements. There are various dimensionally correct methods of determining the mean radius of the droplet based on a polydisperse sample as shown in Equation 11. $\overline{r}_{saut}$ is when p equals 3 and it is useful because it conserves the interfacial area of the droplets.

$$\overline{r}_p = \frac{\sum r_i^p}{\sum r_i^{p-1}} \tag{11}$$

Aerodynamic, optical, and electrical mobility methods exist to measure droplet size. However, due to its low latency, non-invasiveness and direct measurability, Optical Measurements (OM) have gained the most traction (Lin et al., 2024).

OM uses the phenomena of either Rayleigh (RS) and Mie (MS) scattering, where RS is predominantly used, due to its cost advantage and less complexity compared to MS. RS can be used when the droplet circumference is substantially smaller than the wavelength of the light used on OM. Thus, it is important to confirm that droplets for measurement meets that criterion, and foreign particulates are not significantly present, affecting the data, however this cannot be completely known a priori. Uncertainty of early droplet measurements were duly noted, such as particulate effects and some dataset show the final liquid mass fraction or droplet size, rather than multiple data points throughout the back stream (Bakhtar et al., 1995). This lack of data is a point to be scrutinised on literature validating models with these sets. Although some methods try to address the limitations such as conducting sensitivity studies of laser power, effects of filtering steam source. However one should be cautious to survivorship bias (Nagib Elmekawy & Ali, 2020). For instance, if steam sample filtered with a certain mesh gives identical pressure, temperature characteristics with the same steam sample filtered with a finer mesh, this does not imply that heterogenous effects, especially affecting RS data is not removed.

There are other measurement methods that has the potential to support a wide range of droplet sizes and show temporal changes for the smallest of droplets. Schläppi et al (Schläppi et al., 2015) developed a rig that skims backstream nozzle flow and ionises clusters to produces a mass spectra ranging from monomers to 10-20nm. This shows promise to bring insight into early stages of nucleation, how σ is affected in the smallest of clusters. Neutron scattering techniques are also being developed (Wyslouzil et al., 1999), however they are yet to be deployed to an extent comparable to RS.

Environment Geometry

In NEC flow experiments, the boundary layer needs to be considered, due to it decreasing the working area of the blow. This is defined as Boundary Layer Blockage (BLB). Assuming the flow is 1D, the effective area can be inferred by measuring the pressure from a dry expansion. However 2D effects such as pressure undulations,

profile curvature discontinuities and small throat area of nozzle experiments contribute to significant differences in model performances, whether the model assumes the boundary layer of the nozzle is laminar or turbulent (Starzmann et al., 2016).

To add, a major problem arises with replication of benchmark data, when some do not specify exact dimensions of the geometry. In combination with aforementioned issues, this brings further uncertainty in that wish to validate their work.

Dykas et al's cascade (Dykas et al., 2018) is considered to have excellent data, having features such as uncertainties for inlet boundary conditions, qualitative data in the form of Schlieren imagery for all cases, and robust justifications and measurements for water vapour quality (Starzmann et al., 2016).

Table 1: Compilation of NEC Experiment Geometries

Author	Geometry Type	Inlet Boundary Conditions	
		T (K)	P (KPa)
Bakhtar et al (Bakhtar et al., 1995)	Blade Cascade	?	?
Dykas et al (Dykas et al., 2018)	Blade Cascade	369 ~ 380	122 ~103
Moses and Stein (Moses & Stein, 1978a)	Nozzle	255.1 ~ 383.9	3.71 ~ 67.66
Moore (MOORE M., 1973)	Nozzle	?	?
Binne and Green (Binnie & Green, 1942)	Nozzle	374 ~ 426	65.5 ~ 103.4*
Barschdorff (Barschdorff, 1971)	Nozzle	366 ~ 408.7	78.39
Gyarmathy (Gyarmathy, 2005)	Nozzle	560.5 ~ 674.4	208.2 ~ 1497.4
IWSEP (G. Zhang et al., 2021)	Nozzle	374 ~423.5	72.45 ~ 108.8

*originally reported in pounds per square inch

The empty cells in Table 1 represent geometries where the boundary conditions were not found to the best of the author's attempt online and physical sources.

Discussion

Ultimately, experimental data can be used to scrutinise models and evaluate methods that are best practice, whether that is the nucleation, droplet growth or computational model that is involved in the experiment. Metrics such as Mean Square Error (MSE) can be used to assess the simulation data, with parameters such as droplet size and pressure measurements(Lakzian et al., 2023; Starzmann et al., 2018; Y. Zhang & Li, 2026). Furthermore, qualitative phenomena that is observed can be used for visual verification, such as condensation shocks. However, no studies were found that show how much the model matches with the simulation. With quantitative schlieren methods, errors could be quantified regarding this phenomenon (Hargather & Settles, 2012). Difficulty may arise regarding optical quality due to fogging and scattering effects. Current simulation methods and NEC theories has proven itself to show qualitative trends with great accuracy and capture changes within large droplets in the magnitude of several micrometres or higher.

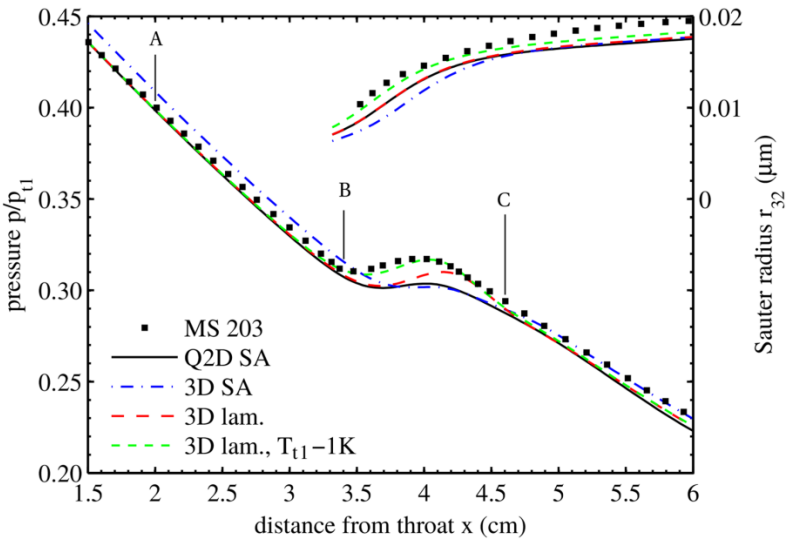


Figure 6: Pressure-Distance diagram of a nozzle experiment. Green dashed line is the modification of a 3D laminar simulation, indicated by the red dashed line, where both are compared with the experimental data, which are the dots(Starzmann et al., 2016).

Uncertainty of the boundary conditions of inlet pressure and temperature is not well established. This is consequential in modelling as shown in Figure 6, where Starzmann

et al (Starzmann et al., 2016) shows that by adjusting the inlet temperature condition by 1K, a significant improve in model accuracy is shown. It is not common practice to indicate explicit error in the data regarding temperature, pressure and radius, which is paramount to validating each model.

In reality, heterogenous effects may arise in the steam and must be considered in systems that have a significant particle concentration. Particles such as NaCl crystals or organic materials trigger this effect. Hu et al created a homogenous-heterogenous coupled model of NEC in OpenFOAM and showed that outlet wetness in a turbine cascade can be controlled by particle radius and concentration, which is the resultant of the particles hindering homogenous NEC, and acting as nucleation cores (Hu et al., 2024).

Although incorporating surface tension is known to improve results, the magnitude of this effect is not well quantified in current literature (G. Zhang, Yang, et al., 2025). Han et al used a Surface Tension Correction Coefficient (STCC) and presented a comparison with a non σ corrected model (Han et al., 2022). STCC is a dimensionless correction on σ_∞, which is another method on capturing σ modelling beyond CNT. By comparing with Moses and Stein (Moses & Stein, 1978b) pressure and liquid mass fraction data, a quadratic fit between inlet superheat and STCC was formulated from 7 'typical' datapoints. This was validated with two other cases of Moses and Stein nozzle. Firstly, a more physically informed fitting is needed inferred from Tolman or BS corrections. Furthermore, by validating the STCC in only one geometry, boundary layer and 3D effects unique to that geometry may apply a bias to the model. Therefore, any models should be validated in a wide range of published data in literature, as shown in Table 1. Although STCC is built from a physical basis, there needs to be further investigation whether errors arising from other factors such as BLB is already extensively considered and the sole point of error may come from the mischaracterisation of σ.

There are numerous studies that test nucleation and droplet growth models. Starzmann et al concluded that due to the complex interactions between droplet

growth and equation of state, the best combination cannot be identified (Starzmann et al., 2018). However, Lazkian et al compared 20 different combinations; 4 nucleation models and 5 droplet growth models that are commonly used and concluded that the J_{HA} with Gyarmathy's growth model had the highest accuracy, based on pressure and droplet radius data from the Bahktar turbine blade (Lakzian et al., 2023). Although the model showed promising accuracy when applied to Moore B and Moore C nozzle data, the inlet conditions both show a similar degree of superheating, which is shown to influence NEC downstream. Therefore, further investigation is needed a wider range of inlet conditions and various geometries to fully validate the combination. Zhang et al compared 7 growth models and addressed a key issue in NEC simulations of inconsistent thermophysical properties. This is the assumption that C_p, the specific heat capacity at constant temperature, stays constant in gaseous phases and liquid phase contributions are ignored, even though it could vary by 20~30% in typical condensation temperatures (Y. Zhang & Li, 2026). Gyamathy's droplet growth model was also concluded to be one of the best performing DGM when tested with IWSEP nozzle pressure and temperature data. However, there is not enough literature that implements J_{HA} that proves superior accuracy. J_{KANT} is commonly used because of its prior implementation on software packages such as Ansys Fluent. J_{CNY} is also rarely utilised, however it is good to note that J_{CNY} correction will influence the results minorly (Bakhtar et al., 2005; Courtney, 1961).

The aforementioned problem regarding experiment geometry is difficult to solve. Results from Dykas shows the best regarding experiments and creating new sets of data with fully defined geometry, clear measurement methods with pressure, liquid mass fraction with further resolution of droplet distribution and inlet conditions is critical. Starzmann et al sums it up succinctly (Starzmann et al., 2016):

"a well-designed nozzle, free from two-dimensional pressure undulations; measurements of centreline pressure distribution in both wet and dry flow; light extinction measurements over a range of wavelengths and at different locations to

obtain reliable droplet size data; and velocity profile measurements at various axial locations, including upstream of the throat."

This literature review shows that although great consideration for accurate measurements has been theorised and segments of it have been practiced, no present study brings the degree of confidence needed for model validation and optimisation. However, simulation that uses an E-E flow field with fine mesh refinements in potential nucleation zones enhanced with MoM for polydisperse tracking, validated by a high-quality dataset and tested with other geometries can be used to test and scrutinise new models, for a more accurate representation of NEC.

Emerging methods to model NEC exists, such as using an artificial neural network to error correct dry gas based models to emulate wet steam (Kang et al., 2025), and advancements with molecular dynamics simulations can be used to further verify models work in the smallest of scales (Graur et al., 2022; Merikanto et al., 2007).

Conclusion

Key pain points to fully define non equilibrium condensation have been discussed, ranging from modelling techniques to experimental methods. It is concluded that modelling using a Eulerian-Eulerian flow field with methods of moments create accessibility to use readily available simulation tools such as Ansys Fluent but also having the ability to simulate phenomena that arises with polydispersity, such as secondary nucleation is critical to get an accurate simulation of non-equilibrium condensation. The importance of high-quality data of pressure and droplet radius measurements have been discussed and Dyka's blade cascade was pointed out to be an exemplar set. Emerging methods for modelling and experimentation have been addressed, such as the use of neural network error correction and ionisation droplet skimming method.

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